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Machine Learning-Based Academic Performance Prediction: A Comparative Empirical Analysis

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Abstract

In this study, we explore the role of various machine learning algorithms in predicting student academic performance from online behavior data while overcoming some limitations posed by traditional subjective assessments. The Decision Tree, Plain Bayes and AdaBoost algorithms were run against the dataset developed from hundreds of observations collected from software engineering students. From the evaluation, it is concluded that prediction accuracy using AdaBoost was 73% which is improved significantly compared to other models. Results thus created give educators some data-driven tools for the early identification of academic risk and optimization of teaching strategies, which extend into a scalable framework for other disciplines in support of improved educational decision-making for successful student outcomes.

CCS Concepts

• **General and reference** → Cross-computing tools and techniques; Experimentation.

Keywords

Academic Performance, Machine Learning, Prediction Algorithm, AdaBoost Algorithm, Experimental Design

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1 Introduction

1.1 Research background

Institutions such as the government and education pay close attention to the academic performance of students at the university level, as good performance defines skill and wielded influence upon future opportunities. Programs in programming stand out as difficult courses in computer science because their complexity and high levels of attrition tend to impede the students' academic progress systematically. Beginning with a move from teacher-centeredness toward more student-centeredness has made the Chinese education system demand more independence from students. Traditional teaching models remain scantily tested and validated in this new paradigm that relies instead on educators' intuition and experience, and therefore have no data-driven basis for establishing optimal outcomes regarding teaching and learning methods, which can negatively impact academic performance [1]. This digital revolution, prodded by the new internet epoch and rapid technological advancements, changed education with methods like educational data mining and learning analytics. Since 2019, machine learning applications in education have risen to use predictive analytics to deliver personalized learning that enhances teaching and learning strategies and improves student outcomes [2,3].

Machine learning is the most important tool for extracting valuable insights from complex datasets in educational data mining, with comparative studies demonstrating that it has an advantage over classical regression methods for predicting student academic performance [3]. The work presented here uses classification algorithms to study the online learning trends in the Fundamentals of Programming (C) course at Sanda University to create a predictive system that optimizes teaching strategies and enhances student learning outcomes [4].

1.2 Research questions

The research seeks to address two main fundamental queries: the predictive accuracies of those three machine learning algorithms in predicting student achievement and the possible identification

for the better performing algorithm on a composite performance indicator. In fact, these research questions now create a context in which to systematically assess the effectiveness of algorithms in predicting educational outcomes:

RQ1: What is the prediction accuracy of the three machine learning methods for predicting student performance?

RQ2: Which algorithm has a better predictive power in terms of predicting student performance?

2 Literature Review

2.1 Student performance prediction

To a large extent, machine learning has redefined the prediction of student performance in higher education, bringing numerous advantages over traditional approaches such as linear regression, which often fail to recognize highly complex patterns in educational data [5]. Support vector machines, decision trees, and neural networks are among the sophisticated algorithms that can automatically learn complex relationships from student data, which in turn yields more accurate predictions of performance and personalized teaching strategies [6]. These methods take into account a wide variety of aspects such as learning attitude, family background, and resource availability into account for the development of individualized learning plans, enhancing students' strengths while working on their weaknesses [7].

Prediction by machine learning algorithms is improving, however, challenges remain in application and interpretation. The problem of feature selection itself is still important for getting the best results; moreover, any data quality problems like missing data value and anomalies can greatly compromise the prediction accuracy [8]. In addition, lack of transparency in the modelling process causes major problems for educators and administrators who need to understand the predictions and trust them before they can be used in practical applications [7]. The ability of machine learning, despite these complications, to revamp educational assessment and facilitate improvement in the quality of teaching and learning will see it be described as a must-have in any modern higher education setting.

2.2 Process-oriented assessment

Process-oriented assessment mashes up learning with an innovative approach on educational assessment, resisting the very establishment of outcome-based methods to assess systematically students' learning experiences. This holistic approach includes classroom participation, group work, and online learning actions in giving teachers insight into students' learning processes and any issues therein. Many times, even these advances can be aided through technological means: online learning platforms become an advanced method, improving upon real time monitoring and data collection about student's progress, personalized advice, and targeted help [9].

Given the aforementioned pedagogically sound nature of the holistic approach of process-oriented assessments towards self-directed learning, while framing the process for such assessments poses quite a few challenges. This includes fairness and impartiality in assessment, managing huge amounts of learner data, and maintaining validity and credibility of the results. However, it could be said that technology has proven itself to be a possible contender for regularly promoting self-awareness and development, thereby reaping better educational dividends via the overall review of the learning process.

2.3 Classification algorithm

Classification algorithms are affecting students' performance forecasting with advanced analytic tools that can identify learning processes and develop targeted interventions [10]. Three algorithms deserve highlighting: Decision trees use an intuitive tree-structured approach, sequentially analyzing features; however, they face difficulty with complex relationships [11]. Simple Bayes applies probabilistic classification quite effectively under the assumption of independence of features but fails if any of the features in question are not independent [12]. The AdaBoost algorithm combines weak classifiers, thus making a stronger one; nevertheless, it is sensitive to noisy data [13].

There has been considerable development due to the introduction of the Proposed Hybrid Classifier (PHC), which is found to perform better on the crucial metrics of accuracy, precision, recall, and F1 score [14]. Other popularly used ones like Random Forests, K-Nearest Neighbors, and Gradient Enhancement Machines have distinct advantages in certain situations, but the choice of algorithms must reflect the needs of the educational environment and context.

2.4 Performance metrics

Performance metrics are important means to quantify the performance of a model on many learning activities. They are different for different kinds of work. For classification tasks, the commonly used metrics include accuracy, precision, recall, F1 score, precision-recall curve (P-R curve), receiver operating characteristic (ROC) and area under the curve (AUC), Matthews correlation coefficient (MCC), and Cohen's kappa. The classification algorithm evaluation measures have been presented in detail by Muntean and Militaru [15,16]. In this study, the algorithms' performance was comparatively analyzed against the set of evaluation metrics.

3 Proposed Solutions

3.1 Research design

The methodology of this study concentrated on the exploration of data concerning the Programming (C) Language course, given for the years 2020-2022 at Shanghai Sanda University, based on data extracted from the Academic Affairs System and the U+ platform [17]. In the pre-processing phase, data was cleansed extensively, and the following statistical analyses were performed: Spearman's rank correlation and tests for normality, arriving at a fine-grained dataset looking mainly at three important assignments and video viewing habits as major indicators of influence over the five semesters [7].

Performance metrics like accuracy, recall, precision, and F1 score were used to evaluate the three chosen algorithms: Simple Bayes, Decision Tree, and AdaBoost. Drawing on these results, an early-warning system for students was developed to classify student

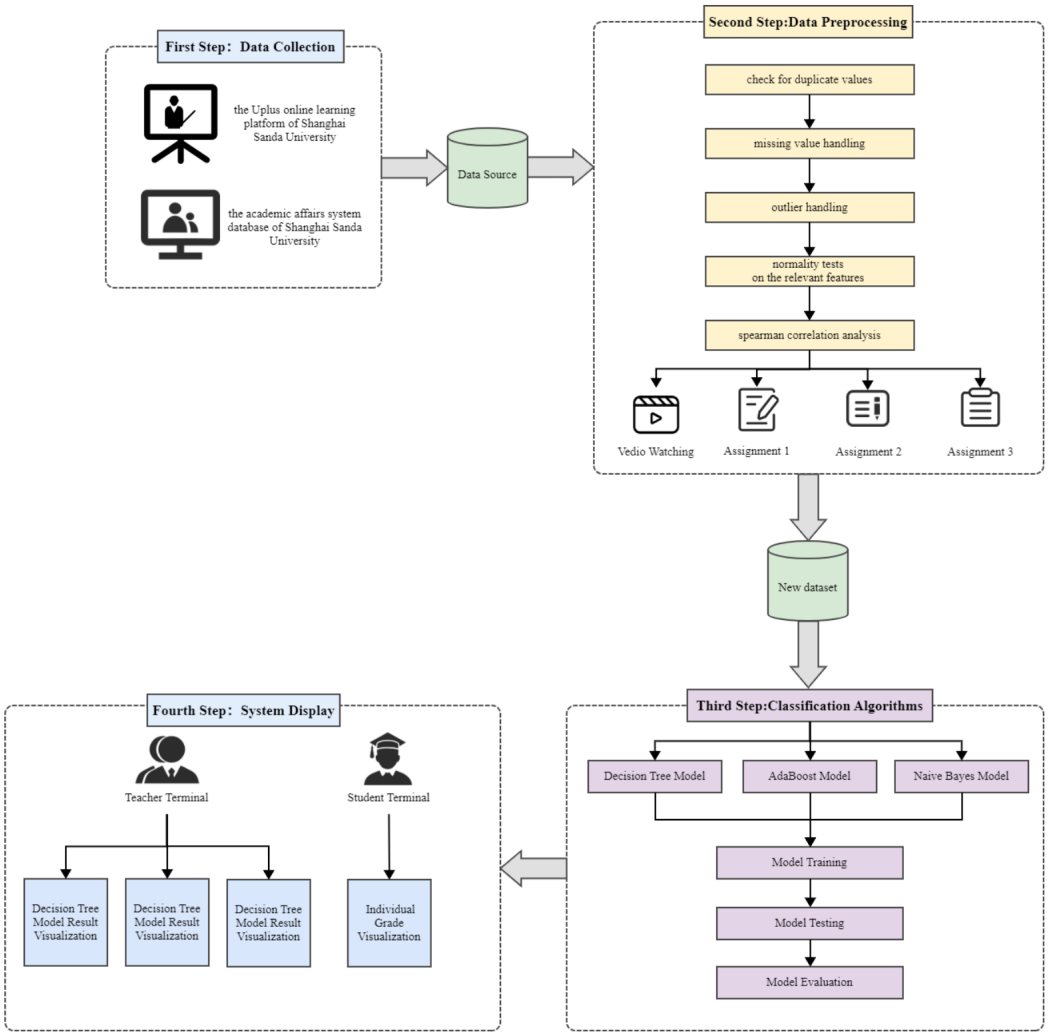

Figure 1: General research design.

performance as either 'good', 'fair', or 'poor', thus giving a timely alert to teachers and students and a clear insight into their progress (Figure 1).

3.2 Data collection and processing

The study dataset was collected between 2020 and 2023 from a 'Programming (C) Language' course in the College of Information Sciences and Technology in Shanghai Sanda University, which covered three classes aggregating a total of 221 students from different semesters (87 students in first semester 2020-2021; 62 and 72 students in the first semester two thousand twenty-two-two thousand twenty-three). Data collection takes into account a variety of information, ranging from the university's information system to its e-learning platform, such as video watching behavior, completion of assignments, and grading.

The results of a thorough correlation study between study behaviors and grade in the final examination are of paramount significance as analyzed through the Shapiro-Wilk test (n=221) and Spearman's correlation coefficient. Findings revealed that all three homework's scored very strongly and positively in correlating with final exam grades (r=0.831, 0.875 and 0.801 respectively, $p<0.001$), while video viewing behavior correlated weakly in the negative direction (r=-0.331, $p<0.001$). The conclusion can be that homework completion is a better measure of academic achievement than video viewing modes (Table 1).

The model-building phase largely revolved around three basic assignment scores from the correlation analysis as the major feature set, while discarding other irrelevant variables, such as student status. Final grades were classified, accordingly, into five grades: A 90-100, B 80-89, C 70-79, D 60-69, and E 0-59-the straightforward classification system to be used for algorithm analysis.

Table 1: Spearman correlation analysis.

Trait	Correlation	P-Value
Video_Watch	-0.331	0.000***
Assignment_Score1	0.831	0.000***
Assignment_Score2	0.875	0.000***
Assignment_Score3	0.801	0.000***

Note: ***, ** and * represent significance levels of 1%, 5% and 10% respectively.

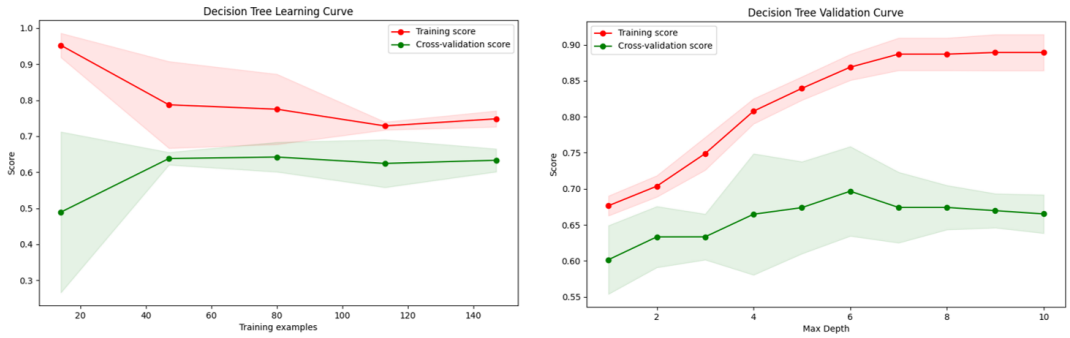

Figure 2: The left one is Decision Tree learning curve; the right one is Decision Tree validation curve.

4 Findings and Discussion

4.1 Testing and evaluation

The effectiveness of the classification models was validated with a variety of performance measures, the most important aspect being accuracy - the ability to make predictions by which one can surmise the performance of a student. Each algorithm was carefully explored with a various range of settings, and results were meticulously documented, taking utmost care to obtain best precision. To further comprehend the models' performance, average performance values were calculated for mean precision, recall, and F1-score derived at each multiplier using the optimal configuration. In addition, to have a complete picture of classification performance, confusion matrices were generated and averaged, thereby informing the prediction patterns and error distributions of the model.

4.2 Decision tree model predicts results

There was a fairly reasonable prediction on the part of the decision tree model regarding the students' performance. The estimated accurateness was 0.62, while precision turned out to be only 0.57. The lesser precision indicated that this model would find it hard to make reliable positive predictions as more of them would be false-positive; it would prove to be much worse in that respect. The balanced F1 measure is also just 0.59, combined with the recall of 0.62, meaning that this model shows limited prediction related to student performance.

The learning curve of the decision tree model reflects how sample size affects the model's performance. The interpretation with 20 training samples is the biggest difference between high training score (0.95) and low cross-validation score (0.5), indicating some overfitting. Now it goes to 110 samples and starts showing relative convergence between training and cross-validation scores, showing generalization but again showing a ceiling in the model's learning capacity.

The validation curves show us how tree depth affects performance. Even if training score increases gradually from 2 (0.67) to 10 with increased depth at the same time, the cross-validation results really recommend that the best generalization should take place at quite shallow or moderate depths, between 6 and 8. This trend thus determines the range of depths at which to optimally balance model complexity and generalization ability; too high, and the risk of under- or overfitting increases (see Figure 2).

4.3 Naive bayes model predicts results

The Naive Bayes classifier is slightly better with prediction accuracies of 0.69 and 0.76 precision in identifying student performance. The model is okay at predicting truly positive events but its recall of 0.69 shows that it could not capture all relevant events. The fair prediction prowess of the model is validated by its balanced F1 of 0.71.

In contrast, the learning curve for the Naive Bayes classifier depicts the relationship between an increasing sample size against a fraction of a sample obtaining lay performance. Initially high training scores, hence near 1.0, indicate overfitting, while convergence toward a similar score between training and cross-validation with increasing samples indicates improvement in generalization. With the training dataset growing, the gap is gradually closing, indicating an increase in stability and reliability of this model.

The validation curves appear to show the influence of 'Var_smooshing' on performance. In the wrong hands, it gave

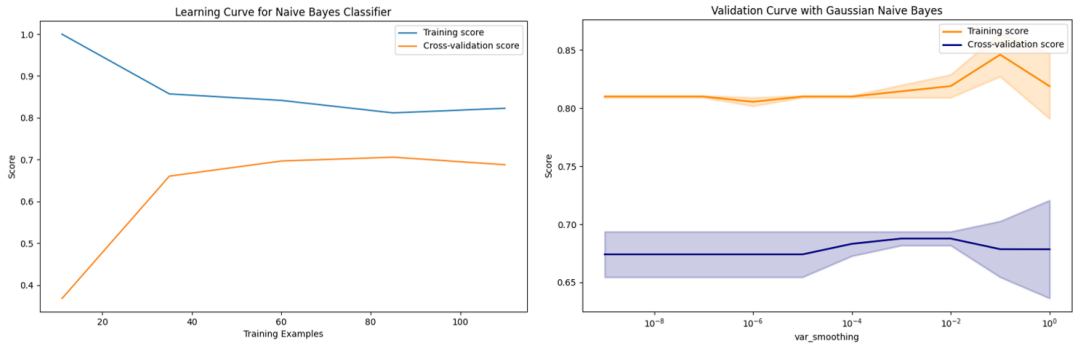

Figure 3: The left one is Learning curve for Naive Bayes Classifier; the right one is Validation curve with Gaussian Naive Bayes.

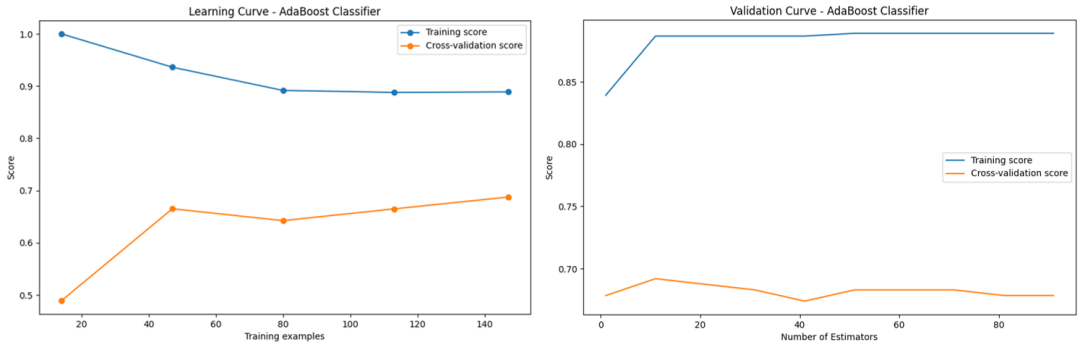

Figure 4: The left one is Learning curve for AdaBoost Classifier; the right one is Validation curve - AdaBoost Classifier.

us too much control over the model: moderate smoothing (10-8) provided a poor cross-validation-generation model, so good in training. At 10-2, Var_smooshing got our model just right between fitting and generalization, whereas cross-validation peaked from 10-4 to 1.0 and leveled off (Figure 3).

4.4 AdaBoost model predicts results

AdaBoost hitherto predicts fairly accurate student performance, showing an accuracy score of 0.73 and precision of 0.80. This model is really good at identifying positive situations, as recalled by a 0.73 recall score, but it cannot capture all relevant events. The model's excellent accuracy-recall trade-off compliance is confirmed by its balanced F1 measure of 0.73, making it a good indicator of student performance.

The learning curve of AdaBoost classifier displays how the model's performance changes as the sample size increases. As the training dataset increases, the convergence of the training and cross-validation scores signifies improved generalization and underlines the criticality of sufficient training loss for model performance. For smaller samples, generally elevated training scores (i.e., around 1.0) in the beginning characterize overfitting.

The validation curves show the relationships between estimator size and model performance. Both training and cross-validation scores peak at around ten estimators before crossing the declining phase-also as the number of estimators increases, the performance of cross-validation faces decline right before attaining the plateau. This model provides a reasonable balance between performance and generalizability of implementation in practice with an appropriate range of 10-20 estimators (Figure 4).

4.5 Results

The purpose of this study was to evaluate the effectiveness of three machine learning algorithms - Decision Tree, Plain Bayes and AdaBoost - in predicting students' performance using metrics including accuracy, precision, recall and F1 measures. The comparative analysis revealed that the Decision Tree algorithm was the worst on both metrics; accuracy was pegged at 0.62, while precision stood at 0.57, well below its competitors. Nevertheless, Simple Bayesian method improves on it with an accuracy score of 0.69 and a precision level of 0.76. However, AdaBoost model improved over it, albeit very modest at precision due to the assumption on the data distribution (Table 2).

Among the various machine models in this study, the AdaBoost model was declared the best performing one, with a class accuracy of 0.73, precision of 0.80, recall of 0.72, and F1 metric at 0.73. Exploiting an ensemble learning model, it was able to efficiently handle complex decision boundaries and nonlinear features. Despite good interpretability, the predictive ability of standard decision trees pales before AdaBoost, hence making AdaBoost the algorithm of choice for predicting student success in educational settings.

Table 2: Comparison of three algorithms.

Algorithm	Accuracy	Prediction	Recall	F1-Measure
Decision tree model	0.62	0.57	0.62	0.59
Naive Bayes model	0.69	0.76	0.69	0.71
AdaBoost model	0.73	0.80	0.73	0.73

5 Conclusion

This project's aims to build a robust and precise model for the prediction of successful students' success or failure in computer science and software engineering that combines extensive data analysis and algorithm selection techniques. Data from Sanda University's IT platform on their learning behavior regarding the "Programming (C) Language" course was investigated to understand student learning characteristics, performance distributions, and influencing factors. This methodology included meticulous preparation and collection of data, followed by careful selection of algorithms and their application. The correlation analysis found that, although there was a strong positive correlation between homework scores and final grades, video viewing time was not included in the prediction model because of low correlation with final grades.

The AdaBoost model turned out to be the best performer with an impressive accuracy, recall, consistency, and interpretability, built over 10 iterations and with the best performance without showing an overfit. Despite issues with accuracy in decision tree model and feature independence assumption of simple Bayesian model, parameter tuning in both models greatly improved the prediction accuracy. This thus makes an advancement in education as it creates a new tool for educators to better understand and predict student performance. The research in feature selection and model design improved predictive accuracy and stability of the models.

Results indicated that these techniques have been deployed in fields requiring precise prediction and data analysis, with implications beyond the education sector. Future studies need to incorporate a larger sample size and more data sources; use sophisticated algorithms for optimization of feature selection and model construction; and investigate new problems within educational data mining areas, such as learning route recommendations and pattern recognition. Other minor limitations, such as being specifically focused on this research and the possibility of inadequate feature selections, warrant further exploration, as doing so may enhance the understanding of student learning and, therefore, provide more meaningful advice and support for teachers.

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